

Benchmark

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships and theorems of triangles.

Benchmark Clarifications:

Clarification 1: Postulates, relationships and theorems include measures of interior angles of a triangle sum to 180° ; measures of a set of exterior angles of a triangle sum to 360° ; triangle inequality theorem; base angles of isosceles triangles are congruent; the segment joining midpoints of two sides of a triangle is parallel to the third side and half the length; the medians of a triangle meet at a point.

Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.

Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.3.1, MA.912.GR.3.2, MA.912.GR.3.3
- MA.912.GR.5.3
- MA.912.T.1.1, MA.912.T.1.2, MA.912.T.1.3, MA.912.T.1.4
- MA.912.LT.4.3, MA.912.LT.4.8, MA.912.LT.4.10

Terms from the K-12 Glossary

- Angle
- Isosceles Triangle
- Triangle

Vertical Alignment

Previous Benchmarks

- MA.8.GR.1.1, MA.8.GR.1.2, MA.8.GR.1.3
- MA.912.AR.2.1

Next Benchmarks

- MA.912.NSO.3
- MA.912.T.1.8

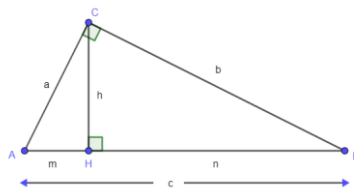
Purpose and Instructional Strategies

In grade 8, students solved problems involving right triangles, including using the Pythagorean Theorem, and angle measures within triangles. In Geometry, students prove relationships and theorems about triangles and solve problems involving triangles, including right triangles. In later courses, students will use vectors and trigonometry to study and prove further relationships between angle measures and side lengths of triangles.

- While the focus of this benchmark are the postulates, relationships and theorems listed in Clarification 1, instruction could include other definitions, postulates, relationships or theorems like a midsegment of a triangle and other angle or side length measures, the Hinge Theorem and the Scalene Triangle Inequality Theorem. Additionally, instruction includes the converse (i.e., if conclusion, then hypothesis) of some postulates and theorems.
- Instruction includes the connection to the Logic and Discrete Theory benchmarks when developing proofs. Additionally, with the construction of proofs, instruction reinforces the Properties of Operations, Equality and Inequality. (MTR.5.1)
- Instruction utilizes different ways students can organize their reasoning by constructing

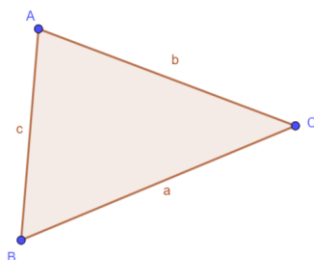
various proofs when proving geometric statements. It is important to explain the terms statements and reasons, their roles in a geometric proof, and how they must correspond to each other. Regardless of the style, a geometric proof is a carefully written argument that begins with known facts, proceeds from there through a series of logical deductions, and ends with the statement you are trying to prove. (MTR.2.1)

- For examples of different types of proofs, please see MA.912.LT.4.8.
- Instruction includes the connection to compass and straight edge constructions and how the validity of the construction is justified by a proof. (MTR.5.1)
- Students should develop an understanding for the difference between a postulate, which is assumed true without a proof, and a theorem, which is a true statement that can be proven. Additionally, students should understand why relationships and theorems can be proven and postulates cannot.
- Instruction includes the use of hatch marks, hash marks, arc marks or tick marks, a form of mathematical notation, to represent segments of equal length or angles of equal measure in diagrams and images.
- Students should understand the difference between congruent and equal. If two segments are congruent (i.e., $\overline{PQ} \cong \overline{MN}$), then they have equivalent lengths (i.e., $PQ = MN$) and the converse is true. If two angles are congruent (i.e., $\angle ABC \cong \angle PQR$), then they have equivalent angle measure (i.e., $m\angle ABC = m\angle PQR$) and the converse is true.
- Instruction includes the use of hands-on manipulatives and geometric software for students to explore relationships, postulates and theorems.
- When proving the Pythagorean Theorem, instruction includes making the connection to similarity criterion.
 - For example, given triangle ABC with $m\angle C = 90^\circ$ and h , the height to the hypotenuse, students can begin the proof that $a^2 + b^2 = c^2$ by first proving that $\triangle AHC \sim \triangle CHB \sim \triangle ACB$ using the Angle-Angle (AA) criterion. Students should be able to conclude that, from the definition of similar triangles, $\frac{a}{c} = \frac{m}{a}$ (in $\triangle AHC \sim \triangle ACB$) and $\frac{b}{c} = \frac{n}{b}$ (in $\triangle CHB \sim \triangle ACB$). Students can equivalently rewrite these equations as $a^2 = cm$ and $b^2 = cn$. Then students can add the resulting equations to create $a^2 + b^2 = cm + cn$. Using algebraic reasoning the segment addition postulate, students can conclude that $a^2 + b^2 = c^2$.

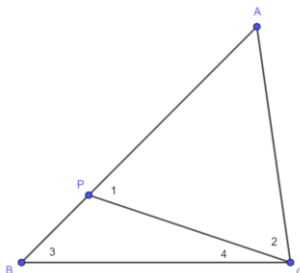


- Instruction includes determining if a triangle can be formed from a given set of sides. The instruction in this benchmark includes discussing the converses of the Triangle Inequality Theorem and the Pythagorean Theorem to understand whether a triangle or a right triangle can be formed from three sides with given side lengths.
 - In triangle ABC , with sides a , b and c , opposite to A , B and C respectively, applying the Triangle Inequality Theorem: $a + b > c$, $b + c > a$ and $a + c > b$. From $a + b > c$, $c < a + b$. From $b + c > a$, $c > a - b$ or $a - b < c$.

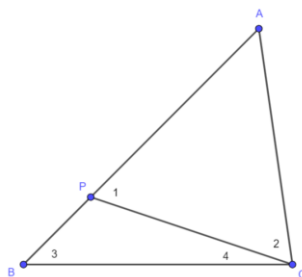
Combining the resulting inequalities: $a - b < c < a + b$.



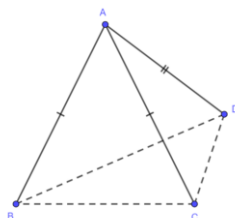
- For example, in a triangle with sides a , b and c , with c being the longest one, if $a^2 + b^2 = c^2$, then the triangle is right. But if $a^2 + b^2 > c^2$, then the triangle is acute, and if $a^2 + b^2 < c^2$, then the triangle is obtuse.
- Instruction includes the understanding of the Scalene Triangle Inequality Theorem (the angle opposite the longer side in a triangle has greater measure), the converse of the Scalene Triangle Inequality Theorem (the side opposite the greater angle in a triangle is longer), and the Hinge Theorem (if two sides of two triangles are congruent and the included angle is different, then the angle that is larger is opposite the longer side).
 - For example, given triangle ABC where $AB > AC$, students can begin the proof of the Scalene Triangle Inequality Theorem by placing the point P such that $AP = AC$. Then, $m\angle 1 = m\angle 2$. As $m\angle ACB > m\angle 2$, $m\angle ACB > m\angle 1$ by the substitution property. $\angle 1$ is exterior to triangle BPC , so $m\angle 1 = m\angle 3 + m\angle 4$ and $m\angle 1 > m\angle 3$. Since $m\angle ACB > m\angle 1$ and $m\angle 1 > m\angle 3$, $m\angle ACB > m\angle 3$ or $m\angle ACB > m\angle ABC$. From this we can conclude, if $AB > AC$ then $m\angle ACB > m\angle ABC$.



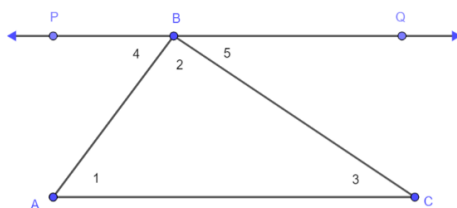
- For example, given triangle ABC where $m\angle ACB > m\angle ABC$ students can prove the converse of the Scalene Triangle Inequality Theorem by using contradiction to show that $AB > AC$. Students can realize that if $AC > AB$, then the Scalene Triangle Inequality Theorem would imply that $m\angle ABC > m\angle ACB$ contradicting the given information. Additionally, students can realize that if $AB = AC$, then $m\angle ABC = m\angle ACB$, which is another contradiction of the given information. Therefore, students can conclude that $AB > AC$.



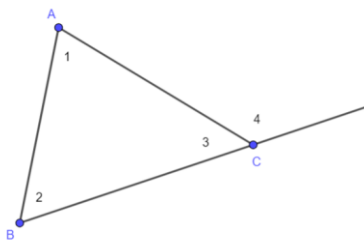
- For example, given triangles ABD and ACB where $AB = AC$ and $m\angle BAD > m\angle CAD$, students can begin the proof of the Hinge Theorem by identifying that triangle ABC is an isosceles triangle with $AB = AC$ and $m\angle ABC = m\angle ACB$. Students should realize that $m\angle DCB > m\angle ACB$, which is the same as $m\angle DCB > m\angle ABC$. Since $m\angle ABC > m\angle DBC$, students can conclude that $m\angle DCB > m\angle DBC$. Applying the Scalene Triangle Inequality Theorem to triangle BCD , students can conclude that $DB > DC$.



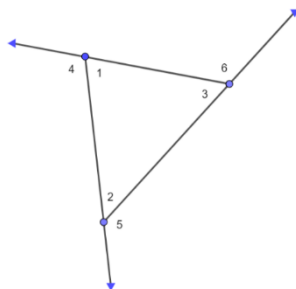
- Instruction includes making the connection to parallel lines and their angle relationships to proving the measures of the interior angles of a triangle sum to 180° .
 - For example, given triangle ABC , students can construct a line through the point B that is parallel AC . Students can then explore the relationships between the angle measures in the image below. Students should realize that $m\angle 4 + m\angle 2 + m\angle 5 = 180^\circ$, and $m\angle 1 = m\angle 4$ and $m\angle 3 = m\angle 5$ (Alternate Interior Angles Theorem). Applying the Substitution Property of Equality, students should be able to conclude that $m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ$.



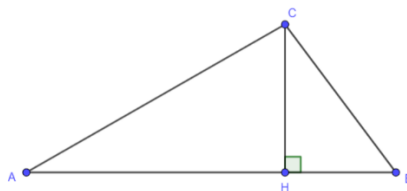
- Instruction includes discussing the relationship between the Triangle Sum Theorem and an exterior angle of a triangle and its two remote interior angles.
 - For example, given triangle ABC , students should realize that $m\angle 4 = m\angle 1 + m\angle 2$.



- Instruction for the proof that the measures of a set of exterior angles of a triangle sum to 360° includes the connection to algebraic reasoning skills, the Triangle Sum Theorem and properties of equality.
 - For example, given the triangle below, students should be able to realize that $m\angle 1 + m\angle 4 + m\angle 2 + m\angle 5 + m\angle 3 + m\angle 6 = 540^\circ$ since there are three pairs of linear pair angles. Applying properties of equalities and the Triangle Sum Theorem, students should be able to conclude that $m\angle 4 + m\angle 5 + m\angle 6 = 360^\circ$.



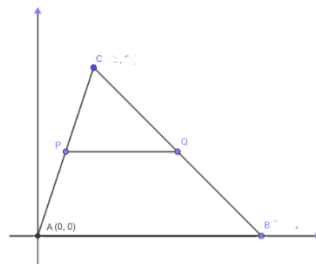
- The proof of the Triangle Inequality Theorem can be approached in a variety of ways. Instruction includes the connection to the Pythagorean Theorem.
 - For example, students can first use the Pythagorean Theorem to prove that the hypotenuse of a right triangle is longer than each of the two legs of the right triangle. Given triangle ABC with an altitude $\overline{CH}$, students can realize that there are two right triangles ACH and BCH ; with $\overline{AC}$ as the hypotenuse of $\triangle ACH$ and $\overline{CB}$ is the hypotenuse of $\triangle BCH$. Students can use their knowledge of right triangles to determine that $AC > AH$ and $AC > HC$, and $CB > HB$ and $CB > CH$. By adding two of the inequalities, $AC > AH$ and $CB > HB$, students can determine that $AC + CB > AH + HB$ which is equivalent to $AC + CB > AB$ by the Segment Addition Postulate.



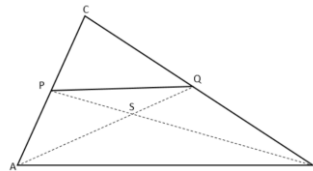
- When proving the Isosceles Triangle Theorem, instruction includes constructing an auxiliary line segment (e.g., median, altitude or angle bisector) from its base to the opposite vertex. (MTR.2.1)
 - For example, given triangle ABC with $\overline{AC} \cong \overline{BC}$, student can construct the median from point C to side AB , with point of intersection M . Students can use the definition of the median of a triangle to state that $\overline{AM} \cong \overline{MB}$. Students should be

able to realize that $\triangle AMC \cong \triangle BMC$ by Side-Side-Side (SSS). So, $\angle A \cong \angle B$ since corresponding parts of congruent triangles are congruent (CPCTC).

- When proving the Triangle Midsegment Theorem, instruction includes the connection to coordinate geometry or to triangle congruence and properties of parallelograms.
 - For example, given triangle ABC on the coordinate plane with A at the origin, B at the point $(b, 0)$ and C at point (x, y) . Students can determine the midpoint of $\overline{AC}$ at the point $P, \left(\frac{1}{2}x, \frac{1}{2}y\right)$ and the midpoint of $\overline{CB}$ at the point $Q, \left(\frac{x+b}{2}, \frac{1}{2}y\right)$. Students should realize that $\overline{PQ}$ is horizontal and is parallel to the base of the triangle, $\overline{AB}$. To determine the length of PQ , students can subtract the x-coordinates of P and Q to find it has a length of $\frac{1}{2}b$. Since $PQ = \frac{1}{2}b$ and $AB = b$, then $PQ = \frac{1}{2}AB$ and that $AB = 2(PQ)$.



- Instruction includes the connection between the Triangle Midsegment Theorem and the Trapezoid Midsegment Theorem. (*MTR.5.1*)
 - For example, students can start with a trapezoid and its midsegment then using geometric software, shrink the top base until it has zero length producing a triangle. Students should be able to realize that the average of the lengths of the two bases of the trapezoids becomes one-half the length of the base of the triangle.
- When proving the medians of a triangle meet in a point, instruction includes the connection to coordinate geometry or to the Midpoint Segment Theorem.
 - For example, given triangle ABC , students can prove that all medians meet at the same point by first constructing two medians, AQ and BP intersecting at the point S . Students should realize that the segment PQ is a midpoint segment. By the Midpoint Segment Theorem, PQ is parallel to AB , therefore students can use the Angle-Angle-Angle (AAA) criterion to state that triangles ASB and QSP are similar. Also, by the Midpoint Segment Theorem, $2PQ = AB$. So the scale factor between the two triangles is 2 and students can conclude that $AS = 2QS$ and that point S is the 2:1 partition point of $\overline{AQ}$.



Students can next look at the median from the point C that intersects $\overline{AB}$ at point R . Using the same procedure as above, students can prove that median $\overline{CR}$ also intersects the median $\overline{AQ}$ at the 2:1 partition point S . Therefore, all three medians

go through the point S .

- For example, students can explore and prove facts about medians of a triangle using the midpoint formula and equations of lines. Students can write the equations of the lines containing two medians and solve the system of equations to determine the point of intersection of the two medians (the centroid). To prove the three medians meet at that point, students can show the centroid is a solution to the equation of the line containing the third median.
- For example, students can use the notion of the weighted average of two points and its connection to the partitioning of line segments to explore and prove facts about medians. The Centroid Theorem states that the centroid partitions the median from the vertex to the midpoint of the opposite side in the ratio 2: 1. In other words, the centroid is $\frac{2}{3}$ of the way from the vertex to the midpoint of the opposite side. This fact about the centroid can also be applied to show how the medians meet at a point.

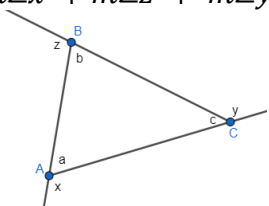
Common Misconceptions or Errors

- Students may extend two sides of a triangle when using exterior angles. An exterior angle of a triangle is formed by the extension of one side of the triangle, not two.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Directions: Print and cut apart the given information, statements and reasons for the proof and provide to students. Students can work individually or in groups. Additionally, students can develop the proof with or without all of the intermediate steps.

Given: $\triangle ABC$ with exterior angles x, y and z. Prove: $m\angle x + m\angle z + m\angle y = 360$. 	
Statements	Reasons
$\triangle ABC$ with exterior angles x , y and z	Given
$\angle a$ and $\angle x$ form a linear pair $\angle b$ and $\angle z$ form a linear pair $\angle c$ and $\angle y$ form a linear pair	Definition of linear pair
$\angle a$ and $\angle x$ are supplementary $\angle b$ and $\angle z$ are supplementary $\angle c$ and $\angle y$ are supplementary	Angles that form a linear pair are supplementary
$m\angle a + m\angle x = 180^\circ$ $m\angle b + m\angle z = 180^\circ$ $m\angle c + m\angle y = 180^\circ$	Definition of supplementary angles